

# ZK-Redact: Efficient Zero-Knowledge Authorized Redaction and Provenance Auditing for Permissioned Blockchains

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**Abstract**—Redactable permissioned blockchains enable legitimate modification of committed transactions, but controlled mutability introduces two important challenges: efficiently verifying that a requester is authorized to perform a redaction and preserving auditable evidence of modifications after the original ledger state has changed. Existing approaches may require disclosure of authorization information, repeated proof verification, or expensive examination of redaction history. This paper proposes ZK-Redact, an efficient privacy-preserving framework for authorized redaction and provenance auditing in permissioned blockchains. ZK-Redact uses zero-knowledge proofs to allow a requester to demonstrate compliance with a registered redaction policy without revealing the private authorization attributes used to satisfy the policy. To reduce processing overhead, authenticated requests are pre-filtered and their independent proofs are processed through a batch-parallel verification workflow before redaction execution. For each successful redaction, ZK-Redact generates an authenticated provenance record binding the target transaction, previous and updated states, applicable policy, and verified authorization evidence. The resulting provenance records are maintained in an authenticated history structure that enables auditors to efficiently verify current transaction integrity and the sequence of authorized redactions using compact cryptographic proofs. ZK-Redact therefore provides privacy-preserving redaction authorization, scalable proof processing, and efficient integrity and provenance auditing within a unified permissioned-blockchain framework.

**Index Terms**—Redactable Blockchain, Zero-Knowledge Proof, Verifiable Redaction, Redaction Provenance, Integrity Auditing, Permissioned Blockchain.

## I. INTRODUCTION

Permissioned blockchains provide shared and tamper-evident records for multi-organization applications. However, strict immutability can conflict with practical requirements such as correcting erroneous data, removing sensitive information, or satisfying privacy regulations. Redactable blockchains address this limitation by enabling authorized modification while preserving ledger consistency, commonly using chameleon hashes (CHs) [1], [2], [9]. Recent studies have extended CH-based redaction with decentralized authorization, fine-grained access control, anonymity, dynamic updates, and verifiability [4], [6], [14]–[16], [20].

Despite these advances, supporting *large-scale* privacy-preserving redaction remains challenging. First, *privacy-preserving authorization can become a verification bottleneck*. Anonymous and privacy-preserving redaction mechanisms protect requester identity or authorization information [4], [14], [17], but independently processing cryptographic

authorization evidence for many concurrent requests increases verification overhead. Existing scalability-oriented schemes mainly optimize blockchain structures or redaction processing [10], [14], [21], leaving scalable privacy-preserving authorization under massive concurrent requests insufficiently addressed.

Second, *redaction execution can become computationally expensive at scale*. CH-based schemes require authorized collision computation and ledger-state updates for each modification [1], [2], [6], [19], while policy enforcement, distributed authorization, and accountability introduce additional processing [3], [15], [20]. In permissioned implementations, individually processing each validated request may further repeat chaincode, transaction, and ledger-update operations. Consequently, large redaction workloads require an efficient mechanism for organizing validated requests and amortizing common blockchain-processing overhead.

Third, *efficient verification of redaction history remains challenging*. Since a valid CH redaction preserves blockchain linkage, the current ledger state alone does not reveal how a transaction has evolved. Existing schemes support traceability, integrity auditing, and authenticated redaction structures [10], [13], [16], [20]; however, verifying multiple successive redactions may still require locating and examining their corresponding historical records. Efficient auditing therefore requires authenticated provenance that supports direct retrieval and compact verification of both individual redactions and the continuity of a transaction’s redaction history.

To address these challenges, we propose **ZK-Redact**, a privacy-preserving framework for scalable redaction and provenance auditing in permissioned blockchains. ZK-Redact uses state- and policy-bound zero-knowledge proofs (ZKPs) for redaction authorization without revealing private credentials or attributes. Logical proof shards distribute ZKP verification across parallel workers, with intra-shard batching when supported. Verified requests are then batch-organized for CH-based redaction, reducing common chaincode and ledger-processing overhead. Finally, transaction-specific hash-linked provenance is Merkle-authenticated and blockchain-anchored, enabling efficient retrieval and independent verification of redaction history.

The main contributions are summarized as follows:

- **Sharded Privacy-Preserving ZK Authorization:** We introduce state- and policy-bound ZK authorization with logical proof sharding for parallel verification. Intra-shard

batching further amortizes verification when supported, reducing ZKP bottlenecks while preserving request-level authorization and failure isolation.

- **Optimized Batch Redaction Processing:** We separate ZKP verification from redaction execution and batch successfully verified requests for processing. State revalidation prevents stale redactions, while common chaincode, validation, and ledger operations are amortized without aggregating request-specific CH computations.
- **Authenticated Provenance and Verifiable Auditing:** We combine transaction-specific hash-linked histories, Merkle-authenticated entries, and blockchain-anchored commitments to support direct provenance retrieval and independent verification of ZKP authorization, history completeness, ordering, and state continuity without exhaustive blockchain traversal.

## II. RELATED WORK

Existing redactable blockchain research has primarily addressed controlled modification, decentralized authorization, privacy, and verifiability. Ateniese *et al.* [9] established the foundation for blockchain rewriting using chameleon hashes (CHs), while Huang *et al.* [2] extended controlled redaction to consortium blockchains for Industrial Internet of Things (IIoT) environments. Subsequent schemes have reduced dependence on a single trapdoor holder through decentralized or distributed CH constructions [1], [6], [12], [19]. Fine-grained redaction has also been investigated through attribute- and policy-based mechanisms [8], [15], [18], while dynamic redaction and trust-aware authorization have been considered in [5], [16]. These schemes strengthen control over *who* can modify blockchain data, but their primary focus is secure redaction rather than optimizing the complete authorization and redaction pipeline under massive concurrent requests.

Privacy and accountability introduce additional requirements. Huang *et al.* [14] proposed scalable redaction with anonymity, while PriChain [4] supports privacy-preserving fine-grained redaction in decentralized settings. More recently, Wang *et al.* [17] considered anonymous and accountable data redaction for IIoT. These approaches demonstrate the importance of protecting requester information while retaining redaction accountability. However, privacy-preserving authorization introduces additional cryptographic processing, and the cited works do not directly address the use of *proof sharding with batch ZKP verification* for distributing large concurrent authorization workloads. ZK-Redact focuses on this scalability dimension by partitioning independently authenticated proofs across logical shards and processing them in parallel and in batches.

Scalability has also been studied from the redaction and data-structure perspectives. Huang *et al.* [14] considered scalable redactable blockchains with update support, while Wu *et al.* [10] introduced an extended Merkle tree for inserted-data redaction in permissioned blockchains. Fugkeaw *et al.* [21] further investigated efficient redaction for large-scale GDPR-compliant blockchain-based e-KYC. Other schemes support dynamic redaction [5], [19], decentralized CH operations

[1], [6], and multi-party CH constructions [12]. Nevertheless, individually executing authorization, CH-related operations, and blockchain updates can still incur repeated processing as the number of redaction requests grows. In contrast, ZK-Redact separates ZKP authorization from redaction execution and organizes already-validated requests for batch-oriented processing, thereby reducing repeated chaincode and ledger-processing operations.

Verifiability and traceability constitute another important research direction. VRBC [13] provides verifiable redaction with efficient query and integrity auditing, while Zhang *et al.* [16] support update and traceability through dynamic trust-based redaction. Miao *et al.* [20] further combine verifiable redaction with lightweight storage and permission supervision. ReAcct [11] considers redaction control in interoperable blockchain environments, and authenticated tree structures such as EMT [10] improve verification of modified blockchain data. These mechanisms provide important foundations for redaction accountability and integrity verification. However, efficiently retrieving and verifying the *successive provenance history of a particular transaction* remains a distinct requirement: an auditor should be able to determine the ordered sequence of redactions and verify continuity between previous and resulting states without exhaustively traversing blockchain records.

In summary, existing work provides strong foundations for controlled CH-based redaction, privacy-preserving authorization, scalable redaction structures, and integrity auditing. ZK-Redact complements these directions by jointly addressing three processing stages that become critical under large-scale workloads: (i) sharded batch ZKP verification for scalable privacy-preserving authorization, (ii) batch-oriented processing of independently validated redactions, and (iii) authenticated provenance organization for efficient retrieval and verification of successive redaction history.

## III. THE PROPOSED ZK-REDACT FRAMEWORK

This section presents the system model, threat model, and construction of ZK-Redact. The framework is designed to efficiently process privacy-preserving redaction requests while preserving verifiable provenance for subsequent auditing.

### A. System Model

ZK-Redact operates over a permissioned blockchain shared by multiple organizations. As shown in Fig. 1, it consists of six components: *Requester (RQ)*, *Redaction Gateway (RG)*, *Proof Verification Layer (PVL)*, *Permissioned Blockchain (PB)*, *Provenance Audit Index (PAI)*, and *Auditor (AU)*.

- 1) **Requester (RQ):** A registered user or organization requesting modification of a blockchain transaction. The RQ constructs the redaction request and generates a zero-knowledge proof (ZKP) demonstrating compliance with the applicable redaction policy without revealing private authorization information.
- 2) **Redaction Gateway (RG):** The RG serves as the admission-control layer. It authenticates and pre-validates incoming requests, filters stale or duplicate requests,

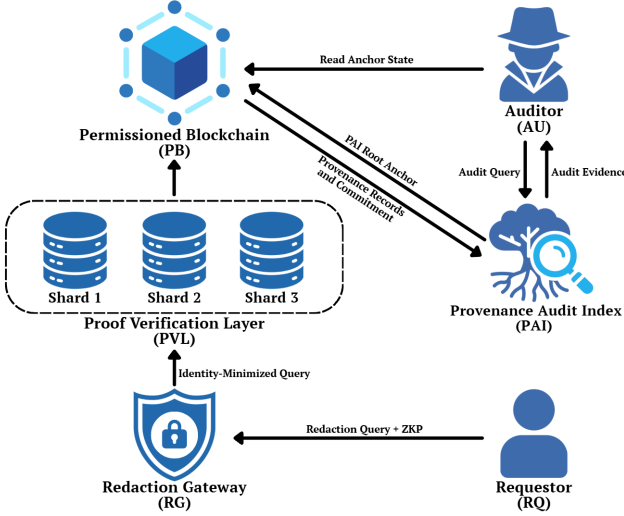


Fig. 1: System model of the proposed ZK-Redact framework

and removes unnecessary requester-identifying information before forwarding accepted requests to the proof-verification layer.

- 3) **Proof Verification Layer (PVL):** The PVL provides scalable ZKP authorization verification. It distributes accepted proofs across logical proof shards for parallel processing and supports intra-shard batched verification when available. Only successfully verified requests proceed to redaction.
- 4) **Permissioned Blockchain (PB):** The PB maintains transactions, redaction policies, and authenticated redaction states. It executes verified redactions using CH-based modification and organizes validated requests for batch-oriented blockchain processing to reduce common chain-code and ledger overhead.
- 5) **Provenance Audit Index (PAI):** The PAI maintains authenticated, transaction-specific redaction histories. It cryptographically links successive redaction states and uses a Merkle-based commitment anchored on the PB to support efficient provenance retrieval and integrity verification.
- 6) **Auditor (AU):** The AU retrieves and independently verifies the provenance of a target transaction. It validates redaction authorization, history integrity and completeness, and successive-state continuity without accessing private ZKP witnesses or traversing unrelated blockchain records.

### B. System Process

ZK-Redact operates through sequential phases covering system setup, privacy-preserving authorization, sharded ZKP verification, batch redaction, provenance maintenance, and audit verification.

**Phase 1: System Setup:** This phase initializes the cryptographic, authorization, proof-processing, blockchain, and provenance components of ZK-Redact.

#### Step 1: System and Entity Initialization

The system administrator initializes

$$PP = (H, \text{Sig}, \text{CH}, \text{ZK}, N, B, \Delta, B_R, \Delta_R) \quad (1)$$

where  $H$  is a collision-resistant hash function;  $\text{Sig}$ ,  $\text{CH}$ , and  $\text{ZK}$  denote the signature, chameleon-hash, and zero-knowledge proof schemes;  $N$  is the number of proof shards; and  $(B, \Delta)$  and  $(B_R, \Delta_R)$  specify the size and waiting bounds for proof-verification and redaction batches, respectively.

Each requester  $U_i$  generates

$$(pk_i, sk_i) \leftarrow \text{Sig.KeyGen}(1^\lambda) \quad (2)$$

and registers  $E_i = (ID_i, pk_i)$ , while retaining  $sk_i$  for request authentication.

Similarly, each auditor  $AU_a$  generates

$$(pk_a, sk_a) \leftarrow \text{Sig.KeyGen}(1^\lambda) \quad (3)$$

and registers

$$E_a^{\text{AU}} = (AID_a, pk_a, \mathcal{P}_a^{\text{audit}}) \quad (4)$$

where  $\mathcal{P}_a^{\text{audit}}$  specifies its permitted audit scope. These registered credentials are subsequently used to authenticate and authorize provenance queries in Phase 6.

#### Step 2: Redaction Policy Registration

For each protected transaction class, the policy administrator defines

$$P_j = (PID_j, \mathcal{A}_j, \mathcal{C}_j, \mathcal{S}_j, V_j) \quad (5)$$

where  $\mathcal{A}_j$ ,  $\mathcal{C}_j$ ,  $\mathcal{S}_j$ , and  $V_j$  specify the authorization predicates, additional conditions, permitted redaction scope, and validity period. Its commitment

$$C_{P_j} = H(PID_j \parallel \mathcal{A}_j \parallel \mathcal{C}_j \parallel \mathcal{S}_j \parallel V_j) \quad (6)$$

is recorded on the blockchain and subsequently bound to the public ZKP statement.

#### Step 3: ZKP and Chameleon-Hash Setup

The system generates

$$(pp_{zk}, vk_{zk}) \leftarrow \text{ZK.Setup}(1^\lambda, \mathcal{R}_{\text{red}}) \quad (7)$$

where  $\mathcal{R}_{\text{red}}$  defines policy-compliant redaction authorization. The proving parameters  $pp_{zk}$  are available to requesters, while  $vk_{zk}$  is provided to the PVL.

The CH parameters are generated as

$$(pk_{ch}, tk_{ch}) \leftarrow \text{CH.KeyGen}(1^\lambda) \quad (8)$$

where  $tk_{ch}$  is protected by the trusted redaction component and used only for authorized CH adaptation.

#### Step 4: Blockchain and Proof-Shard Initialization

The permissioned blockchain deploys chaincode for entity and policy registration, audit-access authorization, transaction updates, redaction results, and provenance commitments. The PVL initializes

$$\mathbb{S} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_N\} \quad (9)$$

where each  $\mathcal{S}_j$  maintains an independent proof queue. Runtime shard assignment and verification-batch formation are performed in Phase 3.

### Step 5: Provenance Audit Index Initialization

For each transaction  $TID_i$  with authenticated initial state  $T_i^{(0)}$ , the PAI initializes

$$v_i = 0, \quad \mathcal{H}_i^{(0)} = \emptyset \quad (10)$$

and computes

$$d_i^{(0)} = H(T_i^{(0)}) \quad (11)$$

$$c_i^{(0)} = H(TID_i \parallel 0 \parallel d_i^{(0)}) \quad (12)$$

The corresponding authenticated entry is

$$A_i^{(0)} = H(TID_i \parallel 0 \parallel c_i^{(0)}) \quad (13)$$

and the initial PAI root is

$$R_{PAI}^{(0)} = \text{MerkleRoot} \left( \left\{ A_i^{(0)} \right\}_{i \in \mathcal{I}} \right) \quad (14)$$

where  $\mathcal{I}$  is the set of indexed transactions.  $R_{PAI}^{(0)}$  is anchored to the permissioned blockchain as the trusted initial provenance state.

**Phase 2: Redaction Request and Privacy-Preserving Authorization:** This phase authenticates each redaction request, proves policy compliance without revealing private authorization attributes, and filters invalid, stale, or duplicate requests before ZKP verification.

### Step 1: Redaction Request and Authentication

For target transaction  $TID_i$  at version  $v_i$ , requester  $U_i$  constructs

$$R_i = (TID_i, v_i, Loc_i, M_i, PID_i, t_i, \nu_i) \quad (15)$$

where  $Loc_i$  and  $M_i$  specify the target location and modification,  $PID_i$  identifies the applicable policy, and  $(t_i, \nu_i)$  provides freshness. The requester authenticates the request by

$$\sigma_i = \text{Sig.Sign}(sk_i, H(R_i)) \quad (16)$$

binding it to the target state, operation, policy, and freshness information.

### Step 2: Policy- and State-Bound ZK Authorization

The requester obtains the current policy version  $v_{P_i}$  and commitment  $C_{P_i}$ , and constructs the public ZKP statement

$$x_i = (TID_i, v_i, Loc_i, M_i, PID_i, v_{P_i} C_{P_i}, t_i, \nu_i) \quad (17)$$

with compact binding digest

$$\eta_i = H(x_i) \quad (18)$$

The private witness is

$$w_i = (Cred_i, \mathcal{A}_i, \rho_i) \quad (19)$$

where  $Cred_i$  is the requester's credential,  $\mathcal{A}_i$  its private authorization attributes, and  $\rho_i$  auxiliary credential evidence.

The authorization relation

$$\mathcal{R}_{\text{red}}(x_i, w_i) = 1 \quad (20)$$

holds iff  $Cred_i$  is valid and non-revoked,  $\mathcal{A}_i$  satisfies the predicates represented by  $C_{P_i}$ ,  $(Loc_i, M_i)$  is within the permitted scope, and the transaction and policy versions and freshness

values are valid. Thus, the proof is bound to the specific redaction request and its governing policy.

The requester generates

$$\pi_i \leftarrow \text{ZK.Prove}(pp_{zk}, x_i, w_i) \quad (21)$$

and submits

$$Q_i = (R_i, v_{P_i}, C_{P_i}, \sigma_i, \pi_i) \quad (22)$$

to the Redaction Gateway (RG).

### Step 3: Gateway Pre-Validation and Deduplication

The RG retrieves the registered  $pk_i$  and verifies

$$\text{Sig.Verify}(pk_i, H(R_i), \sigma_i) = 1 \quad (23)$$

It then validates  $(t_i, \nu_i)$ , confirms that  $v_i$  is the current transaction version, and checks the registered policy tuple  $(PID_i, v_{P_i}, C_{P_i})$ . Invalid or stale requests are rejected before ZKP verification.

For each accepted request, the RG computes

$$DID_i = H(TID_i \parallel v_i \parallel Loc_i \parallel M_i \parallel PID_i) \quad (24)$$

and accepts it only if

$$DID_i \notin \mathcal{P}_{\text{req}} \quad (25)$$

where  $\mathcal{P}_{\text{req}}$  contains pending or previously processed identifiers. The accepted  $DID_i$  is then inserted into  $\mathcal{P}_{\text{req}}$ . Including  $v_i$  prevents duplicate processing against the same transaction state while permitting a new redaction after the state advances.

### Step 4: Identity-Minimized Downstream Submission

After pre-validation, the RG removes the explicit requester identity and constructs

$$Q_i^* = (DID_i, x_i, \eta_i, \pi_i) \quad (26)$$

Since  $x_i$  already contains the transaction, operation, policy, version, and freshness information required for downstream processing, this compact representation avoids duplicating those fields.

The RG securely retains the mapping between  $DID_i$  and the authenticated requester for accountability, while the PVL and downstream components receive only  $Q_i^*$ . Thus, explicit requester identity is hidden from downstream processing without assuming anonymity against the RG.

The accepted  $Q_i^*$  is forwarded to Phase 3 for deterministic proof-shard assignment and parallel ZKP verification.

**Phase 3: Sharded Batch ZKP Verification:** This phase distributes accepted ZK authorization proofs across logical proof shards for parallel verification, combining inter-shard parallelism with intra-shard batched processing.

### Step 1: Proof-Shard Assignment

For each accepted query  $Q_i^*$  from Phase 2, the PVL assigns

$$j_i = 1 + (\text{Int}(H(DID_i)) \bmod N), \quad Q_i^* \rightarrow \mathcal{S}_{j_i} \quad (27)$$

where  $N$  is the number of logical proof shards. The hash-based rule provides uniform workload distribution in expectation. Each shard  $\mathcal{S}_j$  maintains an independent verification queue and is processed concurrently with other active shards.

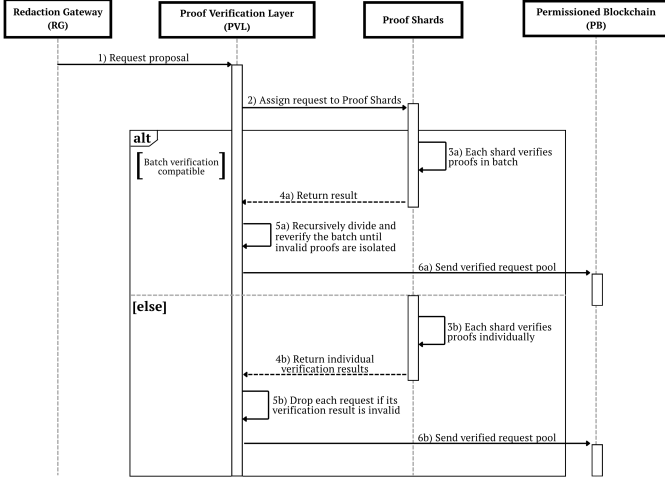


Fig. 2: Sequence of redaction-request verification using distributed proof shards

### Step 2: Shard-Level Batch Formation

Each active shard forms

$$\mathcal{B}_j^{(e)} = \{(x_i, \pi_i, Q_i^*)\}_{i \in I_j^{(e)}}, \quad |\mathcal{B}_j^{(e)}| \leq B \quad (28)$$

where  $I_j^{(e)}$  is the set of requests selected in verification round  $e$ . A batch closes when

$$\text{Close}(\mathcal{B}_j^{(e)}) = (|\mathcal{B}_j^{(e)}| = B) \vee (\tau_j^{(e)} \geq \Delta) \quad (29)$$

where  $\tau_j^{(e)}$  is the waiting time of the oldest request. This permits larger batches under high load while bounding delay under sparse arrivals.

### Step 3: Parallel ZKP Verification

Let  $\mathcal{J}_e$  denote the active shards in round  $e$ . Their batches are processed concurrently, while each request is verified as

$$V_i = \text{ZK.Verify}(vk_{zk}, x_i, \pi_i), \quad V_i \in \{0, 1\} \quad (30)$$

Thus, proof sharding provides parallelism independently of any scheme-specific batch-verification capability.

If the instantiated ZKP scheme supports native batch verification, shard  $\mathcal{S}_j$  may instead compute

$$\mathbf{V}_j^{(e)} \leftarrow \text{ZK.BatchVerify}(vk_{zk}, \{(x_i, \pi_i)\}_{i \in I_j^{(e)}}) \quad (31)$$

where

$$\mathbf{V}_j^{(e)} = \{V_i\}_{i \in I_j^{(e)}} \quad (32)$$

Otherwise, the proofs in  $\mathcal{B}_j^{(e)}$  are independently verified using (30). Hence, native batch verification is an optional optimization rather than a correctness requirement.

### Step 4: Failure Isolation and Verified Request Pool

Verification remains request-specific, and  $Q_i^*$  is accepted only if  $V_i = 1$ . If a native batch verifier returns only an aggregate pass/fail result, a failed batch is recursively divided and reverified until invalid proofs are isolated. This step is unnecessary when per-proof results are available.

Successfully verified requests form

$$\mathcal{P}_{\text{ver}} = \{Q_i^* \mid V_i = 1\} \quad (33)$$

$\mathcal{P}_{\text{ver}}$  is forward to PB while invalid requests are rejected and their pending identifiers are released according to the request-management policy.

Phase 3 therefore uses *proof sharding* for parallel ZKP verification and *intra-shard batching* for efficient scheduling and, when supported, native batch-verification gains.  $\mathcal{P}_{\text{ver}}$  is forwarded to Phase 4 for redaction execution.

**Phase 4: Batch-Verified Redaction and Provenance Generation:** This phase organizes verified requests for batch-oriented processing, executes authorized CH redactions, and generates authenticated provenance evidence. CH adaptation remains request-specific, while common blockchain operations are amortized across the batch.

### Step 1: Redaction Batch Formation

Verified requests in  $\mathcal{P}_{\text{ver}}$  are organized as

$$\mathcal{B}_R^{(e)} = \{Q_i^*\}_{i \in I_R^{(e)}}, \quad |\mathcal{B}_R^{(e)}| \leq B_R \quad (34)$$

where  $B_R$  is the maximum batch size. A batch closes when

$$\text{Close}(\mathcal{B}_R^{(e)}) = (|\mathcal{B}_R^{(e)}| = B_R) \vee (\tau_R^{(e)} \geq \Delta_R) \quad (35)$$

Requests targeting different transactions may execute independently, whereas requests targeting the same transaction are serialized.

### Step 2: State Revalidation and Execution Commitment

Immediately before execution, the PB checks

$$\text{Fresh}_i = [v_i = v_i^{\text{cur}}] \wedge [v_{P_i} = v_{P_i}^{\text{cur}}] \quad (36)$$

where  $v_i^{\text{cur}}$  and  $v_{P_i}^{\text{cur}}$  are the current transaction and policy versions. A stale request is excluded and must be reauthorized. For requests targeting the same transaction, this check is repeated before each transition.

The fresh execution set is

$$I_F^{(e)} = \{i \in I_R^{(e)} : \text{Fresh}_i = 1\} \quad (37)$$

and is bound to the processing round by

$$R_B^{(e)} = \text{MerkleRoot}(\{H(DID_i \parallel \eta_i)\}_{i \in I_F^{(e)}}) \quad (38)$$

where  $\eta_i = H(x_i)$  is the statement-binding digest defined in Phase 2.

### Step 3: Authorized CH Redaction

For each  $i \in I_F^{(e)}$ , let  $m_i^{(v_i)}$  and  $r_i^{(v_i)}$  denote the current content and CH randomness. For modified content  $m_i^{(v_i+1)}$ , the trusted redaction component computes

$$r_i^{(v_i+1)} \leftarrow \text{CH.Adapt}(tk_{\text{ch}}, m_i^{(v_i)}, r_i^{(v_i)}, m_i^{(v_i+1)}) \quad (39)$$

such that

$$\text{CH.Hash}(pk_{\text{ch}}, m_i^{(v_i)}, r_i^{(v_i)}) = \text{CH.Hash}(pk_{\text{ch}}, m_i^{(v_i+1)}, r_i^{(v_i+1)}) \quad (40)$$

The authorized state transition is

$$T_i^{(v_i)} \longrightarrow T_i^{(v_i+1)} \quad (41)$$

Thus, CH adaptation is performed independently, while batching amortizes common chaincode, validation, commitment, and ledger processing.

#### Step 4: Provenance and Audit-Evidence Generation

For each successful redaction, ZK-Redact computes

$$d_i^{\text{old}} = H(T_i^{(v_i)}) \quad (42)$$

$$d_i^{\text{new}} = H(T_i^{(v_i+1)}) \quad (43)$$

and binds its authorization evidence by

$$C_i^{\text{auth}} = H(DID_i \parallel \eta_i \parallel H(\pi_i)) \quad (44)$$

It constructs

$$PR_i^{(v_i+1)} = (TID_i, v_i, v_i + 1, PID_i, v_{P_i}, DID_i, d_i^{\text{old}}, d_i^{\text{new}}, C_i^{\text{auth}}, R_B^{(e)}, t'_i) \quad (45)$$

where  $t'_i$  is the completion timestamp. The public audit evidence  $(x_i, \pi_i)$  is retained off-chain under  $DID_i$  and authenticated by  $C_i^{\text{auth}}$ , enabling independent verification in Phase 6.

#### Step 5: Batch-Result Commitment

Let

$$I_{\text{succ}}^{(e)} \subseteq I_F^{(e)} \quad (46)$$

denote the successfully executed requests. ZK-Redact computes

$$C_B^{(e)} = H \left( R_B^{(e)} \parallel e \parallel \text{MerkleRoot} \left( \left\{ H(PR_i^{(v_i+1)}) \right\}_{i \in I_{\text{succ}}^{(e)}} \right) \right) \quad (47)$$

The PB commits the successful state updates and  $C_B^{(e)}$ ; stale or failed requests are excluded without affecting independent successful redactions. The resulting provenance records are forwarded to the PAI for authenticated maintenance in Phase 5.

Phase 4 therefore separates request-specific CH adaptation from batch-level blockchain processing, preserving per-request correctness while reducing common processing overhead.

**Phase 5: Authenticated Provenance Maintenance:** This phase updates the transaction-specific provenance histories after each completed redaction batch and anchors the resulting PAI state to the blockchain. Processing round  $e$  corresponds to the successful batch-result commitment  $C_B^{(e)}$  generated in Phase 4.

#### Step 1: Provenance-Chain Update

For each successful redaction  $PR_i^{(v_i+1)}$  in round  $e$ , the PAI appends

$$\mathcal{H}_i^{(v_i+1)} = \mathcal{H}_i^{(v_i)} \parallel PR_i^{(v_i+1)} \quad (48)$$

and updates its cumulative commitment as

$$c_i^{(v_i+1)} = H \left( c_i^{(v_i)} \parallel H(PR_i^{(v_i+1)}) \right) \quad (49)$$

where  $c_i^{(0)}$  is initialized in Phase 1. Successive records must satisfy

$$d_i^{\text{new},(v_i)} = d_i^{\text{old},(v_i+1)} \quad (50)$$

thereby binding each redaction output to the next input state.

#### Step 2: Transaction-Specific Index Update

After updating the history, the PAI computes

$$A_i^{(v_i+1)} = H \left( TID_i \parallel (v_i + 1) \parallel c_i^{(v_i+1)} \right) \quad (51)$$

The key  $TID_i$  provides direct access to  $\mathcal{H}_i$ , while the version and cumulative commitment bind its expected length, ordering, and terminal history state.

#### Step 3: PAI Authentication

After all successful records of round  $e$  are incorporated, the PAI computes

$$R_{\text{PAI}}^{(e)} = \text{MerkleRoot} \left( \left\{ A_i^{(v_i)} \right\}_{i \in \mathcal{I}^{(e)}} \right) \quad (52)$$

where  $\mathcal{I}^{(e)}$  is the set of indexed transactions and  $v_i$  denotes each transaction's latest version after round  $e$ . A target entry admits the authentication path

$$\mu_i^{(e)} = \text{MerkleProof} \left( A_i^{(v_i)}, R_{\text{PAI}}^{(e)} \right) \quad (53)$$

#### Step 4: Blockchain Anchoring

After each completed redaction round  $e$ , ZK-Redact computes

$$AC^{(e)} = H \left( e \parallel R_{\text{PAI}}^{(e)} \parallel R_{\text{PAI}}^{(e-1)} \parallel C_B^{(e)} \right) \quad (54)$$

and constructs

$$AR^{(e)} = \left( e, R_{\text{PAI}}^{(e)}, R_{\text{PAI}}^{(e-1)}, C_B^{(e)}, AC^{(e)} \right) \quad (55)$$

The permissioned blockchain records  $AR^{(e)}$  as the trusted provenance anchor. This jointly binds the updated PAI state, its predecessor, and the successful batch-result commitment  $C_B^{(e)}$  to redaction round  $e$ , enabling independent anchor verification in Phase 6.

Phase 5 therefore maintains hash-linked, transaction-specific provenance and authenticates each completed redaction round through a blockchain-anchored PAI root. The authenticated version and terminal commitment enable Phase 6 to detect missing, modified, reordered, or truncated provenance records without scanning unrelated blockchain history.

**Phase 6: Provenance Retrieval and Audit Verification:** This phase enables an authorized auditor to retrieve and independently verify a transaction's redaction history without scanning unrelated blockchain records.

#### Step 1: Auditor Authentication and Query Authorization

Before retrieving provenance, auditor  $AU_a$  constructs

$$R_a^{\text{audit}} = (AID_a, TID_i, e, t_a, \nu_a) \quad (56)$$

where  $AID_a$  is the registered auditor identifier,  $e$  is the target PAI state, and  $(t_a, \nu_a)$  provides freshness. The auditor signs

$$\sigma_a = \text{Sig.Sign}(sk_a, H(R_a^{\text{audit}})) \quad (57)$$

Upon receiving  $(R_a^{\text{audit}}, \sigma_a)$ , the permissioned blockchain verifies

$$\text{Sig.Verify}(pk_a, H(R_a^{\text{audit}}), \sigma_a) = 1 \quad (58)$$

using the registered public key  $pk_a$ , and checks the freshness of  $(t_a, \nu_a)$ . The audit-access chaincode then evaluates

$$\text{AuditAuth}(AID_a, TID_i, e, \mathcal{P}_a^{\text{audit}}) = 1 \quad (59)$$

where  $\mathcal{P}_a^{\text{audit}}$  is the registered audit policy of  $AU_a$ . The query proceeds only if the signature, freshness, and audit-authorization checks are valid; otherwise, it is rejected before provenance retrieval.

### Step 2: Provenance and Audit-Evidence Retrieval

Once authenticated, for target  $TID_i$  and PAI state  $e$ , the auditor submits the retrieval query

$$Q_i^{\text{audit}} = (TID_i, e) \quad (60)$$

The PAI returns

$$\mathcal{E}_i^{(e)} = (\mathcal{H}_i^{(v_i)}, v_i, c_i^{(v_i)}, A_i^{(v_i)}, \mu_i^{(e)}) \quad (61)$$

where  $\mathcal{H}_i^{(v_i)}$  is the ordered provenance history and  $\mu_i^{(e)}$  its Merkle authentication path. Using the  $DID$  in each record, the auditor retrieves the corresponding public evidence  $(x_{i,k}, \pi_{i,k})$  from the off-chain audit-evidence store.

From the permissioned blockchain, the auditor independently obtains the authenticated initial and epoch- $e$  transaction states and the anchor record

$$AR^{(e)} = (e, R_{\text{PAI}}^{(e)}, R_{\text{PAI}}^{(e-1)}, C_B^{(e)}, AC^{(e)}) \quad (62)$$

### Step 3: PAI and History Verification

The auditor recomputes

$$\hat{A}_i^{(v_i)} = H(TID_i \parallel v_i \parallel c_i^{(v_i)}) \quad (63)$$

and verifies

$$\text{MerkleVerify}(\hat{A}_i^{(v_i)}, \mu_i^{(e)}, R_{\text{PAI}}^{(e)}) = 1 \quad (64)$$

Starting from authenticated  $T_i^{(0)}$ , it reconstructs

$$\hat{c}_i^{(0)} = H(TID_i \parallel 0 \parallel H(T_i^{(0)})) \quad (65)$$

and, for  $1 \leq k \leq v_i$ ,

$$\hat{c}_i^{(k)} = H(\hat{c}_i^{(k-1)} \parallel H(PR_i^{(k)})) \quad (66)$$

History integrity and completeness require

$$|\mathcal{H}_i^{(v_i)}| = v_i, \quad \hat{c}_i^{(v_i)} = c_i^{(v_i)} \quad (67)$$

Thus, missing, inserted, modified, reordered, or truncated records cannot match the authenticated PAI entry.

### Step 4: Authorization and State-Continuity Verification

For each  $PR_i^{(k)}$ , the auditor computes

$$\eta_{i,k} = H(x_{i,k}) \quad (68)$$

$$\hat{C}_{i,k}^{\text{auth}} = H(DID_{i,k} \parallel \eta_{i,k} \parallel H(\pi_{i,k})) \quad (69)$$

and accepts its authorization only if

$$\hat{C}_{i,k}^{\text{auth}} = C_{i,k}^{\text{auth}} \quad \wedge \quad \text{ZK.Verify}(vk_{zk}, x_{i,k}, \pi_{i,k}) = 1 \quad (70)$$

The auditor also checks the transaction and policy versions recorded in  $PR_i^{(k)}$ .

For successive redactions, the auditor verifies

$$d_i^{\text{new},(k)} = d_i^{\text{old},(k+1)}, \quad 1 \leq k < v_i \quad (71)$$

and

$$v_i^{(k+1)} = v_i^{(k)} + 1, \quad 1 \leq k < v_i \quad (72)$$

It further verifies the boundary states as

$$d_i^{\text{old},(1)} = H(T_i^{(0)}) \quad (73)$$

$$d_i^{\text{new},(v_i)} = H(T_i^{(v_i,e)}) \quad (74)$$

where  $T_i^{(0)}$  is the authenticated initial state and  $T_i^{(v_i,e)}$  is the authenticated transaction state associated with the audited PAI state  $e$ .

### Step 5: Anchor Verification

Finally, the auditor verifies

$$AC^{(e)} \stackrel{?}{=} H(e \parallel R_{\text{PAI}}^{(e)} \parallel R_{\text{PAI}}^{(e-1)} \parallel C_B^{(e)}) \quad (75)$$

This binds the authenticated PAI state to its predecessor and the successful batch-result commitment of redaction round  $e$ .

The audit succeeds only if the PAI membership, provenance commitment, ZKP authorization, state continuity, and blockchain anchor are valid. Thus, ZK-Redact supports transaction-specific verification of redaction authorization, history integrity, completeness, and continuity without exhaustive blockchain traversal.

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